

Variables Manipulation: Java

Casting

Casting is the process of transforming a value of one primitive data type to another.

A `double` value can be transformed into an `int` value using the `(int)` casting operator, while an `int` value can be transformed into a `double` value using the `(double)` casting operator.

```
// Convert double to int
double numDouble = 12.99;
System.out.println((int) numDouble); //
Prints: 12

// Convert int to double
int numInt = 9;
System.out.println((double) numInt); //
Prints: 9.0
```

Math Operations

Basic math operations can be applied to `int`, `double` and `float` data types:

- `+` addition
- `-` subtraction
- `*` multiplication
- `/` division
- `%` modulo (yields the remainder)

These operations are not supported for other data types.

```
int a = 20;
int b = 10;

int result;

result = a + b; // 30

result = a - b; // 10

result = a * b; // 200

result = a / b; // 2

result = a % b; // 0
```

Comparison Operators

Comparison operators can be used to compare two values:

- `>` greater than
- `<` less than
- `>=` greater than or equal to
- `<=` less than or equal to
- `==` equal to
- `!=` not equal to

They are supported for primitive data types and the result of a comparison is a boolean value `true` or `false`.

```
int a = 5;
int b = 3;

boolean result = a > b;
// result now holds the boolean value true
```

Compound Assignment Operators

Compound assignment operators can be used to change and reassign the value of a variable using one line of code. Compound assignment operators include `+=`, `-=`, `*=`, `/=`, and `%=`.

```
int number = 5;

number += 3; // Value is now 8
number -= 4; // Value is now 4
number *= 6; // Value is now 24
number /= 2; // Value is now 12
number %= 7; // Value is now 5
```

Increment and Decrement Operators

The increment operator, `(++)`, can increase the value of a number-based variable by `1` while the decrement operator, `(--)`, can decrease the value of a variable by `1`.

```
int numApples = 5;
numApples++; // Value is now 6

int numOranges = 5;
numOranges--; // Value is now 4
```

Order of Operations

The order in which an expression with multiple operators is evaluated is determined by the order of operations:

parentheses → multiplication → division → modulo →
addition → subtraction.

 **Print**